

Shaft Analysis

This file provides the relevant computations for validating the selection and dimensions of components like the shaft and bearings. This includes evaluating tendon-induced loads on pulleys and shafts, friction effects, bearing reactions, and estimations of stress and bearing-life.

Coordinate Frame & Geometry

The global coordinate frame is defined as:

- x : proximal $\rightarrow$ distal
- y : (lateral) left $\rightarrow$ right
- z : palmar $\rightarrow$ dorsal

Pulley positions and shaft axes are defined in the global frame for vector-based computation.

Each pulley instance is defined as:

- center : c
- shaft axis : $\hat{v}$
- radius : r
- friction coefficient : μ

Capstan Equation

For a cable tensioned around a cylinder with friction, the holding force exerted on one side is related to the loading force on the other by the capstan equation. Applied tension $T_{load}(\phi)$ is a function of the total angle subtended by the rope on the capstan (wrap angle ϕ) and frictional force.

$$T_{load}(\phi) = T_{hold}e^{\mu\phi}$$

Tangent-Based Wrap

To find the (incoming or outgoing) tangent line from a tendon wrapped around a pulley, we first project the vector v between the two pulley centers into the plane normal to the shaft axis.

$$v_{plane} = v - (v \cdot \hat{a})\hat{a}$$

The tangent angle offset is

$$\alpha = \sin^{-1}\left(\frac{r}{d}\right)$$

Where r is the radius of the current (internal) pulley and d is the projected distance between the two pulley centers. The tangent direction then becomes

$$t = \cos(\alpha u) + s \cdot \sin(\hat{a} \times u)$$

Where $s = \pm 1$ determines the routing side (tangent vector direction).

The wrap angle is then calculated as

$$\theta = \angle(t_{in}, t_{out})$$

Tension Propagation

Tension along the tendon is modeled using the capstan equation to account for friction.

T_{out}	friction model
T_{in}	no friction
$T_{in}e^{-\mu\theta}$	energy loss
$T_{in}e^{+\mu\theta}$	amplification

Pulley Force Calculation

The force on each internal pulley is

$$F = T_{in}t_{in} + T_{out}t_{out}$$

Shaft Reaction Forces

Each shaft is modeled as a simply supported beam with two bearings.

- L : bearing span
- F_i : transverse load at position s_i

Force equilibrium for the bearing reactions and applied loads is

$$R_1 + R_2 = \sum F_i$$

Moment equilibrium about bearing 1:

$$R_2 L = \sum F_i s_i$$

$$R_1 = \sum F_i - R_2$$

Bearing Loads and Life

The radial and axial loads on a bearing from the bearing reaction is

$$F_a = |R \cdot \hat{a}|$$

$$F_r = ||R - F_a \hat{a}||$$

The dynamic equivalent radial load for a bearing is expressed as

$$P = XF_r + YF_a$$

Where X and Y are load factors dependent on the bearing and can be found from bearing tables.

The basic rating life is then

$$L_{10} = \left(\frac{C}{P}\right)^p 10^6$$

Where $p = 3$ for ball bearings and C is the dynamic load rating.